Number 9 — Comparative costs · Taxation · Salaries and income tax · Foreign currency

Calculator allowed. Marks in brackets. ★ = harder question. Money answers to the nearest cent/penny unless told otherwise.

1 Which car has the better fuel efficiency? Show your working. [4]

car	fuel used	distance
Vega	9.5 litres	150 km
Orion	13.2 litres	220 km

2 A phone costs £552 including sales tax at 20%. Calculate the price before tax. [3]

3 The country of Numeria has these income-tax rates. [8]

tax rate	annual salary
0%	first \$10 000
20%	\$10 001 to \$40 000
40%	above \$40 000

Mr Costa earns \$36 000 a year and Mrs Costa earns \$58 000 a year. How much tax does the family pay per week?

4 £1 (UK pounds) $=$ \$1.28 (US dollars) $=$ €1.17 (euros). Convert [6]

a £40 into dollars **b** \$40 into pounds **c** €40 into dollars.

5 Nadia wants to buy a flat in Spain costing €240 000 using UK pounds. In March the exchange rate was £1 $=$ €1.15; in June it was £1 $=$ €1.20. How much would she save, in pounds, by buying in June rather than March? [4]

Total: 25 marks

Check: best value = compare per unit (per litre, per 100 g, km per litre) · a price including tax is (100 + rate)% of the original — divide, don't subtract · tax bands: only the slice inside each band is taxed at that rate · currency: multiply to go from the £1 side, divide to come back · convert via the common currency when neither is £.